

Back-of-the-Envelope Estimation

In a **system design interview**, you are often asked to estimate **capacity, scale, traffic, storage, bandwidth, latency, or server count** quickly.

This is called **back-of-the-envelope estimation**.

It means using rough assumptions + common system numbers to judge whether a design is feasible.

1. What It Means

Instead of exact calculations:

None

Use approximate numbers

Use rounded values

Use logic + engineering intuition

Goal is not perfection.

Goal is:

None

Understand scale

Choose correct architecture

Avoid impossible designs

Show engineering thinking

2. Why Interviewers Ask This

They want to test if you understand:

None

Traffic scale

Storage growth

Memory needs

Bandwidth usage

Database load

Caching impact

Server capacity

3. Example Questions

Typical interview prompts:

None

How many servers needed for WhatsApp?

How much storage for YouTube videos?

How many QPS for Instagram likes?

Can one DB handle traffic?

How much Redis memory needed?

4. Common Estimation Areas

A. Traffic

None

Daily Active Users

Requests per second

Peak traffic

Reads vs writes

B. Storage

None

User data

Logs

Images

Videos

Backups

C. Memory

None

Cache size

Session storage

Hot data

D. Compute

None

Servers needed

Workers needed

CPU capacity

5. Basic Formula Set

Requests Per Second

None

$$\text{QPS} = \text{Total Daily Requests} / 86400$$

Peak QPS

None

$$\text{Peak} = \text{Average} \times 3 \text{ to } 10$$

Storage Per Year

None

$$\text{Daily Data} \times 365$$

Bandwidth

None

$\text{QPS} \times \text{Response Size}$

6. Example 1: Social Media Likes

Suppose:

None

10 million users

Each user likes 5 posts/day

Total likes/day:

None

50 million writes/day

Average writes/sec:

None

$50,000,000 / 86400 \approx 580 \text{ writes/sec}$

Peak:

None

$\approx 2,000 \text{ writes/sec}$

Now you know one DB may or may not handle it.

7. Example 2: URL Shortener Storage

Suppose:

None

100 million new URLs/year

Each row = 500 bytes

Storage:

None

$100,000,000 \times 500$

= 50 GB/year

Very manageable.

8. Example 3: Video Platform

Suppose:

None

1 million uploads/day

Each video = 100 MB

Daily storage:

None

100 TB/day

Need object storage + CDN.

9. Common Assumptions Used in Interviews

None

1 KB = 10^3 bytes

1 MB = 10^6 bytes

1 GB = 10^9 bytes

1 day = 86,400 sec

Peak traffic = 3x to 10x average

Read traffic > write traffic in most apps

10. What Interviewer Wants to Hear

Speak like this:

None

Assume 10M DAU

Each user makes 20 requests/day

That gives 200M requests/day

≈ 2,300 RPS average

Peak maybe 10K RPS

This shows structured thinking.

11. Important Rule

Use **reasonable assumptions** and clearly state them.

Even if assumption differs slightly, that is okay.

Bad answer:

None

I don't know exact number.

Good answer:

None

I'll assume 5 million users and estimate from there.

12. Where It Helps in Design

Estimation helps choose:

None

SQL vs NoSQL

Single DB vs sharding

Need CDN or not

Need cache or not

Need Kafka or not

Need async workers or not

13. Interview Tip

Always estimate:

None

Users

QPS

Storage

Peak load

Growth over 1–3 years

Before deep architecture discussion.

14. One-Line Summary

Back-of-the-envelope estimation is a quick approximate calculation method used in system design interviews to evaluate traffic, storage, and scalability needs using logical assumptions and simple math.